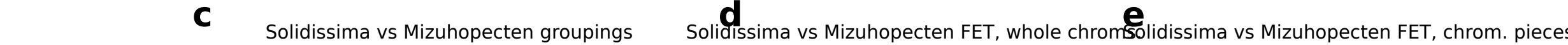


Solidissima vs Mizuhopecten synteny (chromosome coordinates)



	Solidissima scaf	Mizuhopecten scaf	p (FET)	Count
0	Chr13 : 0-end	NW_018404492.1 : 0-end	6.18e-87	82
1	Chr10 : 0-end	NW_018408896.1 : 0-end	3.59e-68	60
2	Chr05 : 0-end	NW_018408902.1 : 0-end	7.03e-57	58
3	Chr01 : 0-end	NW_018405386.1 : 0-end	1.08e-54	58
4	Chr13 : 0-end	NW_018407390.1 : 0-end	5.70e-57	55
5	Chr02 : 0-end	NW_018404940.1 : 0-end	2.32e-57	52
6	Chr05 : 0-end	NW_018480077.1 : 0-end	5.45e-46	50
7	Chr04 : 0-end	NW_018406397.1 : 0-end	1.23e-50	48
8	Chr05 : 0-end	NW_018407065.1 : 0-end	2.67e-43	43
9	Chr10 : 0-end	NW_018404097.1 : 0-end	6.38e-45	41
10	Chr14 : 0-end	NW_018408465.1 : 0-end	1.43e-39	40
11	Chr04 : 0-end	NW_018407700.1 : 0-end	2.22e-36	40
12	Chr10 : 0-end	NW_018408713.1 : 0-end	2.23e-42	39
13	Chr10 : 0-end	NW_018407435.1 : 0-end	2.23e-42	39
14	Chr08 : 0-end	NW_018408653.1 : 0-end	3.78e-38	38
15	Chr07 : 0-end	NW_018408629.1 : 0-end	4.10e-35	38
16	Chr03 : 0-end	NW_018405944.1 : 0-end	1.25e-29	38
17	Chr06 : 0-end	NW_018408899.1 : 0-end	2.60e-35	37
18	Chr05 : 0-end	NW_018404662.1 : 0-end	1.51e-29	37
19	Chr07 : 0-end	NW_018408761.1 : 0-end	5.77e-36	36
20	Chr12 : 0-end	NW_018408252.1 : 0-end	3.17e-35	35
21	Chr06 : 0-end	NW_018404373.1 : 0-end	7.85e-36	35
22	Chr06 : 0-end	NW_018403543.1 : 0-end	4.83e-31	35
23	Chr01 : 0-end	NW_018407509.1 : 0-end	3.64e-30	35
24	Chr03 : 0-end	NW_018408638.1 : 0-end	1.06e-32	34
25	Chr11 : 0-end	NW_018404269.1 : 0-end	9.99e-37	33
26	Chr06 : 0-end	NW_018480076.1 : 0-end	4.77e-32	33
27	Chr17 : 0-end	NW_018408880.1 : 0-end	1.07e-32	32
28	Chr14 : 0-end	NW_018404481.1 : 0-end	6.55e-37	32
29	Chr10 : 0-end	NW_018406415.1 : 0-end	6.29e-29	32
30	Chr08 : 0-end	NW_018480078.1 : 0-end	4.15e-30	32
31	Chr03 : 0-end	NW_018408887.1 : 0-end	4.70e-29	32
32	Chr12 : 0-end	NW_018408742.1 : 0-end	2.71e-36	31
33	Chr07 : 0-end	NW_018408154.1 : 0-end	5.60e-27	31
34	Chr11 : 0-end	NW_018408916.1 : 0-end	8.20e-27	30
35	Chr06 : 0-end	NW_018408645.1 : 0-end	1.71e-27	30
36	Chr05 : 0-end	NW_018406089.1 : 0-end	3.09e-27	30
37	Chr03 : 0-end	NW_018408895.1 : 0-end	9.44e-26	30
38	Chr14 : 0-end	NW_018405900.1 : 0-end	1.33e-31	29
39	Chr12 : 0-end	NW_018408921.1 : 0-end	2.36e-30	29
40	Chr04 : 0-end	NW_018406060.1 : 0-end	3.98e-27	29
41	Chr02 : 0-end	NW_018408919.1 : 0-end	7.29e-30	29
42	Chr14 : 0-end	NW_018405342.1 : 0-end	3.50e-29	28
43	Chr12 : 0-end	NW_018403589.1 : 0-end	3.05e-31	28
44	Chr02 : 0-end	NW_018407364.1 : 0-end	1.25e-28	28
45	Chr07 : 0-end	NW_018404519.1 : 0-end	1.40e-29	27
46	Chr06 : 0-end	NW_018408770.1 : 0-end	1.04e-26	27
47	Chr06 : 0-end	NW_018406690.1 : 0-end	1.46e-21	27
48	Chr05 : 0-end	NW_018405747.1 : 0-end	1.11e-24	27
49	Chr04 : 0-end	NW_018408914.1 : 0-end	8.21e-25	27
50	Chr03 : 0-end	NW_018408913.1 : 0-end	3.91e-25	27
51	Chr15 : 0-end	NW_018408726.1 : 0-end	6.40e-29	26
52	Chr12 : 0-end	NW_018407426.1 : 0-end	1.82e-31	26
53	Chr08 : 0-end	NW_018408623.1 : 0-end	9.13e-27	26
54	Chr05 : 0-end	NW_018407907.1 : 0-end	1.73e-24	26
55	Chr05 : 0-end	NW_018404308.1 : 0-end	1.32e-25	26
56	Chr04 : 0-end	NW_018405113.1 : 0-end	5.25e-19	26
57	Chr01 : 0-end	NW_018406268.1 : 0-end	8.27e-23	26
58	Chr13 : 0-end	NW_018406874.1 : 0-end	2.30e-26	25
59	Chr06 : 0-end	NW_018407733.1 : 0-end	5.98e-22	25
60	Chr03 : 0-end	NW_018407511.1 : 0-end	5.62e-23	25
61	Chr16 : 0-end	NW_018408820.1 : 0-end	1.23e-25	24
62	Chr09 : 0-end	NW_018408786.1 : 0-end	1.35e-21	24
63	Chr08 : 0-end	NW_018403786.1 : 0-end	2.54e-24	24

(a) depicts the chromosomes/scaffolds of *Solidissima* (x-axis) plotted against the chromosomes/scaffolds of *Mizuhopecten* (y-axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10858 orthologs found between 24 *Solidissima* scaffolds and 1556 *Mizuhopecten* scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visually finding gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not provide information on which gene group is correlated with the chromosome p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.